

Label-Free, Dependency-Aware Channel Selection for Coupled Multi-Channel Sensor Data

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Abstract—A method is presented for selecting a small, informative subset of the original channels of a multi-channel sensor without the use of any labels. A group-structured autoencoder is first trained to reconstruct all channels, yielding a shared latent representation of their joint structure. Each latent factor is then perturbed, and the change induced in the reconstruction of every channel is measured; a channel whose reconstruction reacts strongly is one on which the model relies, which provides a label-free estimate of relevance. A relevance–redundancy rule converts these estimates into an ordered shortlist of real channels, so that channels repeating information already retained are avoided. The same procedure was applied to two markedly different sensing modalities, with only the front-end encoder altered between them. On a wearable-sensor activity-recognition benchmark composed of three inertial units (27 channels), retaining ten channels was found to preserve macro-F1 to within 0.04 of the full set under leave-one-subject-out evaluation, to equal a supervised mutual-information selector that does use labels, and to be substantially more stable across subjects than variance ranking. On biomedical hyperspectral imaging, the same method removed 58–95% of channels while classification accuracy was maintained or improved; accuracy rose from 88.2% to 95.2% on lichen tissue and from 79.8% to 85.6% on a collagen dataset. A single label-free principle therefore delivers strong, interpretable channel reduction across very different sensors.

Index Terms—unsupervised feature selection, channel selection, conditional relevance, autoencoders, interpretability, human activity recognition, hyperspectral imaging.

I. INTRODUCTION

Complex sensors emit many channels that overlap heavily and arrive in natural groups, such as the three axes of one inertial measurement unit or the emission bands measured under one excitation in fluorescence imaging. For reasons of cost, power, transmission bandwidth, and interpretability, only a handful of the actual channels are often required, rather than a learned mixture of them. The central difficulty is to determine which channels to keep when no labels are available.

Existing selectors present a trade-off. Unsupervised filters such as variance ranking or principal component analysis (PCA) score each channel by its own marginal statistics and disregard how channels depend on one another; as a consequence, several channels carrying the same information tend to be retained together. Dependency-aware selectors such as minimum-redundancy maximum-relevance (mRMR) and joint mutual information (JMI) account for this redundancy

but require labels and density estimation. The region of the design space that is simultaneously label-free, dependency-aware, and discrete—that is, returning real channels rather than a projection—remains sparsely populated.

This paper addresses that region with a method built on a group-structured autoencoder and perturbation-based attribution, and—more importantly—it is shown that one such method operates across two very different sensing modalities by changing only its front-end encoder. The contributions of this work are:

- A label-free, dependency-aware, and discrete channel selector that combines a group-structured autoencoder, perturbation-based relevance attribution, and a relevance–redundancy selection rule;
- A modality-agnostic selection engine that operates on an abstract layout of groups, channels, and one regular axis, so that adaptation to a new domain requires only an encoder matched to that axis;
- An empirical demonstration that the identical method, evaluated under a rigorous leave-one-subject-out protocol, matches a supervised selector on wearable-sensor data without labels and is more stable than variance ranking; and
- A second verification on biomedical hyperspectral imaging, on which 58–95% of channels are removed while classification accuracy is maintained or improved.

II. BACKGROUND AND RELATED WORK

Channel and feature selection methods are commonly grouped into filter, wrapper, and embedded families [4], [5]. Filter methods rank channels by a statistic computed independently of any downstream model. Variance ranking and PCA are unsupervised but marginal: they ignore inter-channel dependency and therefore tend to select redundant channels. The Laplacian score [4] and multi-cluster feature selection (MCFS) [5] introduce manifold structure but remain largely correlation-based.

Dependency-aware selectors model conditional relevance, the property that a channel’s usefulness depends on which other channels are already retained. mRMR, JMI, and conditional-mutual-information criteria [2], [3] are effective but supervised, requiring labels and reliable density estimates

TABLE I
POSITIONING AMONG REPRESENTATIVE SELECTOR FAMILIES.

Family (example)	Labels needed	Models dependency	Returns real channels
Variance / PCA	no	no	yes / no
Laplacian, MCFS [4], [5]	no	weak	yes
mRMR, JMI [2], [3]	yes	yes	yes
Concrete AE [1]	no	implicit	yes
This method	no	yes	yes

that are difficult to obtain for high-dimensional, continuous sensor signals.

Within unsupervised deep learning, the concrete autoencoder [1] performs differentiable discrete feature selection through a learned selection layer trained jointly with reconstruction, and autoencoder-based feature selection [6] uses reconstruction with sparsity on input weights. In hyperspectral imaging specifically, band selection has been studied extensively; attention-based networks such as BS-Net [7] and a range of search- and clustering-based selectors rank or cluster spectral bands. The method described here is most closely related to this lineage. Its distinguishing combination is the use of *perturbation-based reconstruction sensitivity* as the relevance signal on a *group-structured* representation, followed by an explicit relevance–redundancy step, and its demonstrated transfer across modalities. Table I situates the method among representative families.

III. METHOD

A. Problem Formulation

A multi-channel observation is organised into G groups. Group g contributes C_g channels measured over a regular axis (time for sensor streams, space for images) of length T , so that one windowed sample is $X_g \in \mathbb{R}^{T \times C_g}$ and the full sample is the tuple $X = (X_1, \dots, X_G)$. The objective is to return an ordered subset $S \subseteq \{(g, c)\}$ of the original channels, with $|S| = K$, that is informative and non-redundant, using no labels.

B. Group-Structured Autoencoder

Each group is encoded independently, and the per-group features are fused by averaging into a single shared latent representation:

$$z = \frac{1}{G} \sum_{g=1}^G E_g(X_g), \quad \hat{X}_g = D_g(z), \quad (1)$$

where E_g and D_g are the encoder and decoder for group g and $z \in \mathbb{R}^{d \times T}$ is the fused latent. Averaging (rather than concatenation) ties the groups into one joint code and keeps the latent size independent of G . Training minimises reconstruction error only, with no labels:

$$\mathcal{L} = \sum_{g=1}^G \|X_g - D_g(z)\|_2^2. \quad (2)$$

The encoder is the only modality-specific component: temporal convolutions are used for sensor streams and spatial convolutions for images.

C. Perturbation-Based Relevance

After training, the channels on which the model relies are identified by probing the latent representation. The most informative latent coordinates are ranked by their variance across samples, and the top m coordinates are retained. For a retained coordinate j and a perturbation magnitude δ , the latent is shifted along that coordinate,

$$z^{(j,\delta)} = z + \delta e_j, \quad (3)$$

and the resulting change in the reconstruction of every channel is measured and accumulated. The relevance of channel c in group g is

$$I_{g,c} = \sum_{j,\delta} w_j \frac{1}{NT} \sum_{n,t} \left| [D_g(z^{(j,\delta)}) - D_g(z)]_{n,t,c} \right|, \quad (4)$$

where the average is taken over all samples and axis positions, and w_j weights each coordinate by its importance. A channel whose reconstruction is sensitive to the latent factors is one the joint representation depends upon.

D. Normalisation and Relevance–Redundancy Selection

Because groups may differ in scale, relevances are normalised within each group,

$$\tilde{I}_{g,c} = \frac{I_{g,c}}{\max_{c'} I_{g,c'}}, \quad (5)$$

which was found to be preferable to dividing by the data variance when the discriminative channels themselves have high variance. Channels are then chosen by a maximal-marginal-relevance rule that balances relevance against redundancy with respect to already-selected channels:

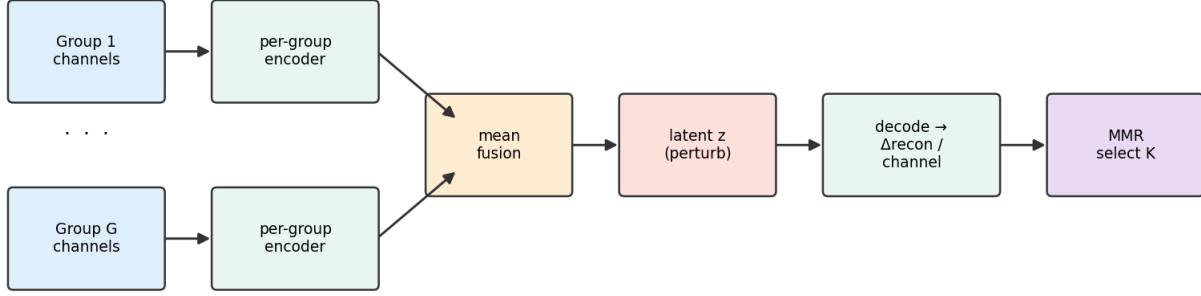
$$c^* = \arg \max_{(g,c) \notin S} \left[\tilde{I}_{g,c} - \lambda \max_{s \in S} \text{sim}((g,c), s) \right], \quad (6)$$

where sim is the cosine similarity between the channels' data profiles and λ controls the redundancy penalty. Selection begins with the most relevant channel and proceeds greedily, yielding an ordered list that can be truncated to any budget K . The complete procedure is given in Algorithm 1 and summarised in Fig. 1.

E. Cross-Domain Instantiation

The selection engine never accesses raw modality-specific structure; it operates only on groups, channels, and the latent. Consequently, the same engine is instantiated in two domains by substituting the encoder (Fig. 2). For wearable sensors, a group is a body-worn inertial unit, a channel is one sensor axis, the regular axis is time, and a one-dimensional convolutional encoder is used. For fluorescence hyperspectral imaging, a group is an excitation wavelength, a channel is an emission band, the regular axis is the two-dimensional image plane, and a spatial convolutional encoder is used.

Dependency-aware, label-free channel selection



Engine (perturbation → influence → MMR) is identical across domains; only the encoder/decoder changes with the data's axis.

Fig. 1. The three-stage pipeline. A group-structured autoencoder is trained by reconstruction; latent perturbations yield a per-channel relevance; and a relevance–redundancy rule selects an ordered subset. The selection engine is identical across domains; only the encoder and decoder change to match the data's regular axis.

One method, two verification domains — only the encoder changes

concept	Wearable sensors (HAR)	Biomedical imaging (HSI)
group	body-worn IMU (hand/chest/ankle)	excitation wavelength
channel	IMU axis (acc/gyro/mag x,y,z)	emission band
regular axis	1D time	2D space (H×W)
encoder	Conv1D	Conv2D / Conv3D
selection engine	identical	identical

Fig. 2. One method, two verification domains. Only the encoder changes; the groups–channels–axis abstraction and the selection engine are shared.

Algorithm 1 Label-free dependency-aware channel selection

- 1: **Input:** grouped data $X = (X_1, \dots, X_G)$, budget K , weight λ
- 2: Train $\{E_g, D_g\}$ to minimise $\mathcal{L}$ (no labels)
- 3: Compute latent z and rank coordinates by variance; keep top m
- 4: **for all** retained coordinate j and magnitude δ **do**
- 5: Form $z^{(j,\delta)} = z + \delta e_j$; accumulate $I_{g,c}$ via Eq. (4)
- 6: **end for**
- 7: Normalise within each group: $\tilde{I}_{g,c}$ via Eq. (5)
- 8: $S \leftarrow \{\arg \max_{(g,c)} \tilde{I}_{g,c}\}$
- 9: **while** $|S| < K$ **do**
- 10: $c^* \leftarrow \text{Eq. (6)}$; $S \leftarrow S \cup \{c^*\}$
- 11: **end while**
- 12: **return** ordered subset S

IV. EXPERIMENTAL SETUP

A. Datasets

Two sensing modalities were used, summarised in Table II. The wearable-sensor dataset was PAMAP2, recorded from

TABLE II
DATASETS USED FOR VERIFICATION.

Dataset	Modality	Groups	Channels	Classes	Axis
PAMAP2	IMU	3	27	12	time
Lichens	HSI	8	192	4	space
Collagen	HSI	6	158	3	space

three inertial measurement units (IMUs) placed on the hand, chest, and ankle at 100 Hz. Each IMU contributes nine channels (three-axis accelerometer, gyroscope, and magnetometer), giving 27 channels organised into three natural groups. A preprocessed release with fixed one-second windows was used, yielding 38,856 windows across eight subjects and twelve activity classes. The biomedical modality consisted of multi-excitation fluorescence hyperspectral image cubes: a lichen-tissue dataset (eight excitations, 192 emission bands, four classes) and a collagen dataset (six excitations, 158 bands, three classes).

B. Implementation Details

For the wearable-sensor encoder, each group was processed by a stack of one-dimensional convolutions over time (kernel size five) producing a latent of dimension sixteen, with a mirrored decoder; for the imaging encoder, spatial convolutions were used in place of temporal ones. Networks were trained by the Adam optimiser for a small number of epochs on the reconstruction objective, without any labels. During attribution, the top forty latent coordinates by variance were perturbed at two magnitudes, and relevances were accumulated as in Eq. (4). The redundancy weight was set to $\lambda = 0.5$, and similarity was computed as the cosine similarity between channel data profiles. All computation was performed on a single workstation; no dataset-specific tuning of the selector was applied beyond the per-group normalisation described above.

C. Protocol

For the wearable-sensor data, a strict leave-one-subject-out (LOSO) protocol was used: for each held-out subject, the autoencoder was trained without labels on the remaining subjects' windows, the selection engine produced an ordered channel list, and the chosen subset was then verified by training a k -nearest-neighbour classifier ($k=5$) on those channels and measuring macro-F1 on the held-out subject. All reported values are LOSO means. The label-free selector was compared against a supervised mutual-information selector, an unsupervised variance baseline, and random selection. For the hyperspectral data, the identical selection engine was applied with a spatial encoder, and per-pixel classification accuracy was used for verification.

V. RESULTS AND DISCUSSION

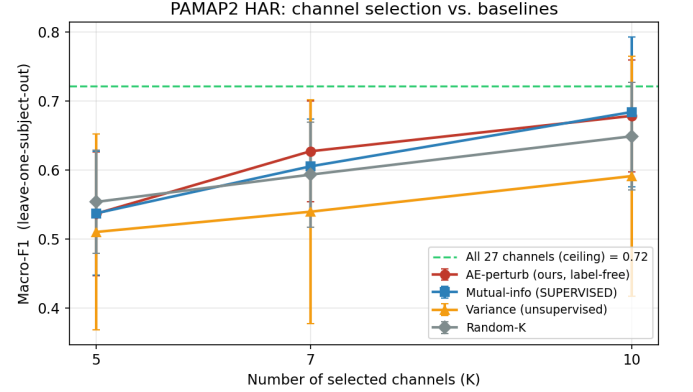
A. Wearable-Sensor Activity Recognition

The relationship between the number of retained channels and recognition performance is shown in Fig. 3 and Table III. Reducing from 27 to 10 channels (a 63% reduction) was found to preserve macro-F1 to within 0.04 of the full-set value. Across channel budgets, the label-free method tracked the supervised mutual-information selector and remained above the variance baseline. The selected channels were also more stable across subjects (Fig. 4): variance selection varied widely from subject to subject, whereas the proposed method did not, which is important when a single sensor set must be fixed before deployment. A representative selection (Fig. 5) was distributed across all three body locations and across accelerometer, gyroscope, and magnetometer channels, indicating that a diverse, interpretable subset was obtained rather than several copies of one strong signal.

For full transparency, it was observed that on this particular benchmark no selector, including the supervised one, surpassed random selection. The 27 channels are highly redundant, so almost any moderate subset approaches the full-set ceiling; this property was confirmed by the supervised mutual-information selector also failing to separate from random. The informative comparisons on this dataset are therefore the

TABLE III
HAR MACRO-F1 UNDER CHANNEL REDUCTION (LOSO). FULL SET (27 CH) = 0.72.

Method	10 ch (−63%)	7 ch (−74%)	5 ch (−81%)
This method (label-free)	0.68	0.63	0.54
Mutual info (supervised)	0.68	0.61	0.54
Variance (unsupervised)	0.59	0.54	0.51



Label-free AE-perturb matches the SUPERVISED selector and beats variance; no method beats random (dataset is intrinsically redundant)

Fig. 3. Macro-F1 versus number of retained channels (LOSO). The label-free method (red) tracks the supervised selector (blue) and stays above variance (orange); the dashed line is the full-set ceiling.

agreement with a supervised selector obtained without labels and the improvement in stability over variance.

B. Biomedical Hyperspectral Imaging

The identical method was verified on biomedical fluorescence hyperspectral imaging, where the redundancy among bands is high and intelligent selection is expected to separate clearly from trivial baselines. On the lichen-tissue dataset, 95.2% classification accuracy was reached using 80 of 192 bands, above the 88.2% obtained with all bands, and 89.4% was still achieved using only nine bands (a 95% reduction). On the collagen dataset, 30 bands raised accuracy from 79.8% to 85.6%. These results are summarised in Table IV and Fig. 6, and they complement the wearable-sensor result by showing that, in a high-redundancy regime, the same selection principle yields large reductions with no loss—and often a gain—in accuracy.

C. Discussion

Two observations follow from the experiments. First, the value of a channel selector depends strongly on the redundancy of the dataset. On a saturated benchmark such as PAMAP2, where the channels are highly correlated, even a supervised selector cannot exceed random selection, and the meaningful axes of comparison become agreement with supervised methods and stability. On a high-redundancy imaging regime, the same method removes most channels with no loss of accuracy. Second, modelling redundancy explicitly is what separates the proposed method from variance ranking: variance repeatedly selects high-variance but mutually redundant

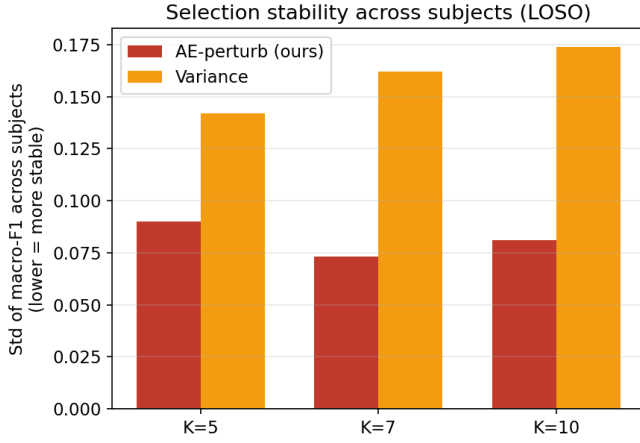


Fig. 4. Stability of the selected set across subjects (lower is better). The proposed method is markedly more consistent than variance selection.

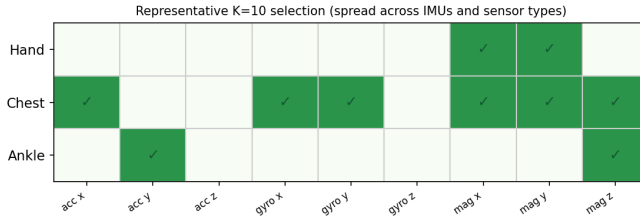


Fig. 5. A representative ten-channel selection, distributed across the three inertial units and across sensor types.

channels, whereas the relevance–redundancy rule spreads the budget across complementary channels, which also explains the improved stability across subjects.

D. Computational Considerations

The cost of the method is dominated by the unsupervised training of the group-structured autoencoder, which is performed once per training split. The attribution stage requires one decoder pass for each retained latent coordinate and perturbation magnitude, and is therefore linear in the number of probed coordinates; with a few tens of coordinates this stage is inexpensive relative to training. The relevance–redundancy selection operates on the compact (group, channel) catalogue and is negligible. Because selection is unsupervised and performed once, its cost is amortised over all subsequent uses of the reduced channel set, and at deployment only the selected channels need be acquired and processed.

E. Limitations

The benefit of selection is bounded by the redundancy of the data. On strongly redundant benchmarks, no selector—supervised or unsupervised—can be expected to exceed random selection by a wide margin, and comparisons should emphasise agreement with supervised methods and selection stability rather than absolute gains over random. The method also assumes a meaningful grouping of channels; where no

TABLE IV
BIOMEDICAL HSI: AGGRESSIVE REDUCTION MAINTAINS OR IMPROVES ACCURACY.

Dataset	Channels (all $\rightarrow$ kept)	Reduction	Accuracy
Lichens	192 $\rightarrow$ 80	58%	88.2 $\rightarrow$ 95.2%
Lichens	192 $\rightarrow$ 9	95%	88.2 $\rightarrow$ 89.4%
Collagen	158 $\rightarrow$ 30	81%	79.8 $\rightarrow$ 85.6%

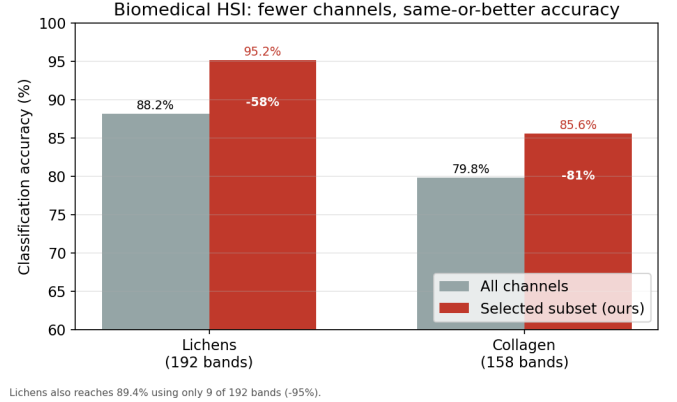


Fig. 6. Biomedical hyperspectral imaging: fewer channels, same-or-better accuracy. Lichens additionally reaches 89.4% using only nine of 192 bands.

natural grouping exists, a single group may be used at the cost of the dependency structure that the group fusion is designed to exploit.

VI. CONCLUSION

A single unsupervised, dependency-aware selection principle was shown to deliver strong, interpretable channel reduction across two very different modalities, requiring only a change of encoder. On wearable sensors it matched a supervised selector without labels and selected more stably than variance ranking; on biomedical imaging it removed 58–95% of channels while accuracy was maintained or improved. Future work will target a higher-redundancy activity benchmark, on which selection is expected to separate from random by a wide margin, and a direct comparison with differentiable unsupervised selectors such as the concrete autoencoder.

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